

| Question                            |                        |                    |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks available |                   |                    |                                     |                        |      |       |         |   |                  |         |   |  |   |  |   |  |  |
|-------------------------------------|------------------------|--------------------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------------------|--------------------|-------------------------------------|------------------------|------|-------|---------|---|------------------|---------|---|--|---|--|---|--|--|
|                                     |                        |                    |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | AO1             | AO2               | AO3                | Total                               | Maths                  | Prac |       |         |   |                  |         |   |  |   |  |   |  |  |
|                                     | (d)                    |                    |  | 7.87 g of the amino acid gives 1470 cm <sup>3</sup> of N <sub>2</sub> gas (1)<br><br>n(N <sub>2</sub> ) = 1470 / 24.5 × 1000 = 0.060<br><br>therefore 0.060 mol of the amino acid<br><br>molar mass of the amino acid = 7.87 / 0.060 = 131 (1)<br><br>R—CH(NH <sub>2</sub> )COOH → 131 and CH(NH <sub>2</sub> )COOH → 74<br><br>therefore R → 131 – 74 = 57 (1)<br><br>it is a straight chain compound so R is CH <sub>3</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>2</sub><br><br>therefore amino acid must be CH <sub>3</sub> (CH <sub>2</sub> ) <sub>3</sub> CH(NH <sub>2</sub> )COOH (1) |                 | 2                 |                    |                                     | 2                      |      |       |         |   |                  |         |   |  |   |  |   |  |  |
|                                     | (e)                    | (i)                |  | 2-methylpropane-1,3-dioic acid<br><br>accept 2-methylpropanedioic acid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1               |                   |                    | 1                                   |                        |      |       |         |   |                  |         |   |  |   |  |   |  |  |
|                                     |                        | (ii)               |  | <table border="1"><thead><tr><th>Protons</th><th>Splitting pattern</th><th>Relative peak area</th></tr></thead><tbody><tr><td>–CH<sub>2</sub>–CH<sub>2</sub>–</td><td>singlet / no splitting</td><td>4</td></tr><tr><td>–C–H–</td><td>quartet</td><td>1</td></tr><tr><td>–CH<sub>3</sub></td><td>doublet</td><td>3</td></tr></tbody></table><br>award (1) for each correct row                                                                                                                                                                                                                  | Protons         | Splitting pattern | Relative peak area | –CH <sub>2</sub> –CH <sub>2</sub> – | singlet / no splitting | 4    | –C–H– | quartet | 1 | –CH <sub>3</sub> | doublet | 3 |  | 3 |  | 3 |  |  |
| Protons                             | Splitting pattern      | Relative peak area |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                 |                   |                    |                                     |                        |      |       |         |   |                  |         |   |  |   |  |   |  |  |
| –CH <sub>2</sub> –CH <sub>2</sub> – | singlet / no splitting | 4                  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                 |                   |                    |                                     |                        |      |       |         |   |                  |         |   |  |   |  |   |  |  |
| –C–H–                               | quartet                | 1                  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                 |                   |                    |                                     |                        |      |       |         |   |                  |         |   |  |   |  |   |  |  |
| –CH <sub>3</sub>                    | doublet                | 3                  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                 |                   |                    |                                     |                        |      |       |         |   |                  |         |   |  |   |  |   |  |  |
|                                     |                        |                    |  | Question 10 total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3               | 8                 | 3                  | 14                                  | 2                      | 1    |       |         |   |                  |         |   |  |   |  |   |  |  |